



47. Let D_n be the $n \times n$ matrix with $d_{ij} = 1$ if $i = j$ and $d_{ij} = 0$ if $i \neq j$. Let A_n be the $n \times n$ matrix with $a_{ij} = 1$ if $i = j$ and $a_{ij} = 0$ if $i \neq j$.

[illegible]